

redOx 2015

Trends in Redox Flow Battery Technology and project REDOX2015

International Workshop on Energy Storage in the grid.

Barcelona, 9 January 2014

Luis Santos

Agenda

1. EDP in Spain

2. Conventional approach to storage: price arbitrage

3. New approach: grid services and reserves

4. Redox flow batteries are a promising option

5. Redox2015 project

6. Conclusions



1. EDP in Spain



**#1
Portugal**

**First Portuguese company by market
capitalisation**

**#1
Europe**

**~ 3 GW of hydraulic projects under
construction**

**#3
Worldwile**

**More than 6.4 GW of wind
capacity**

**4º Electricity producer and
Distribution System Operator**

2º Operator of natural gas



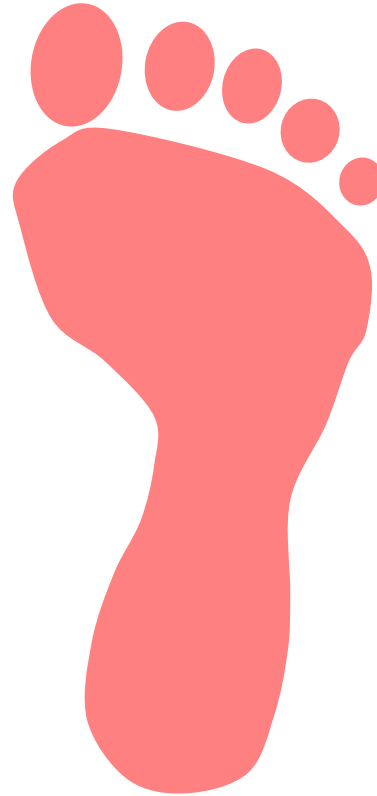
1. EDP in Spain

Innovation strategic lines

**Flexible generation and
with high availability**



**Environmental
innovation**



Electric mobility



Grids and storage



Energy intelligence



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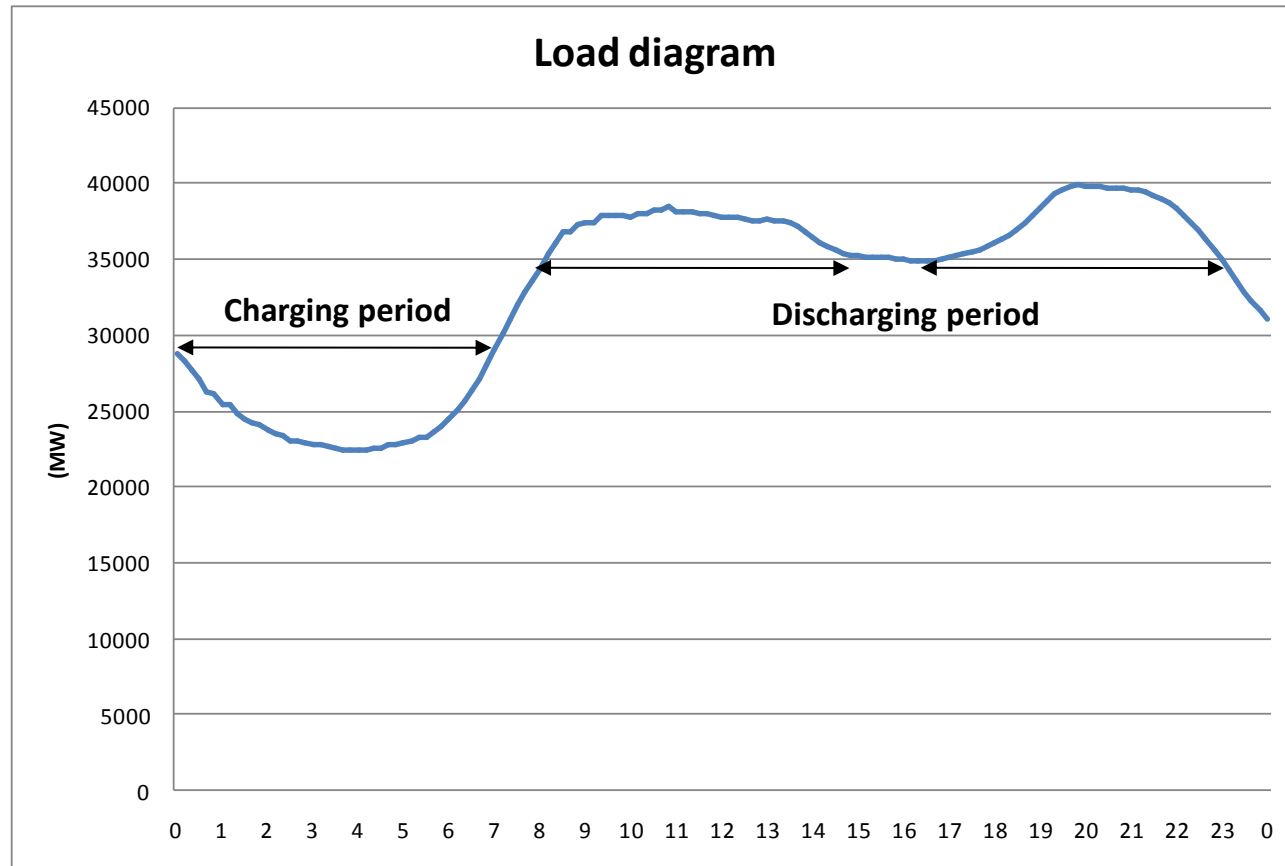
6. Conclusions



2. Conventional approach to storage: price arbitrage

As a first approach, the economics of energy storage are assessed by price arbitrage

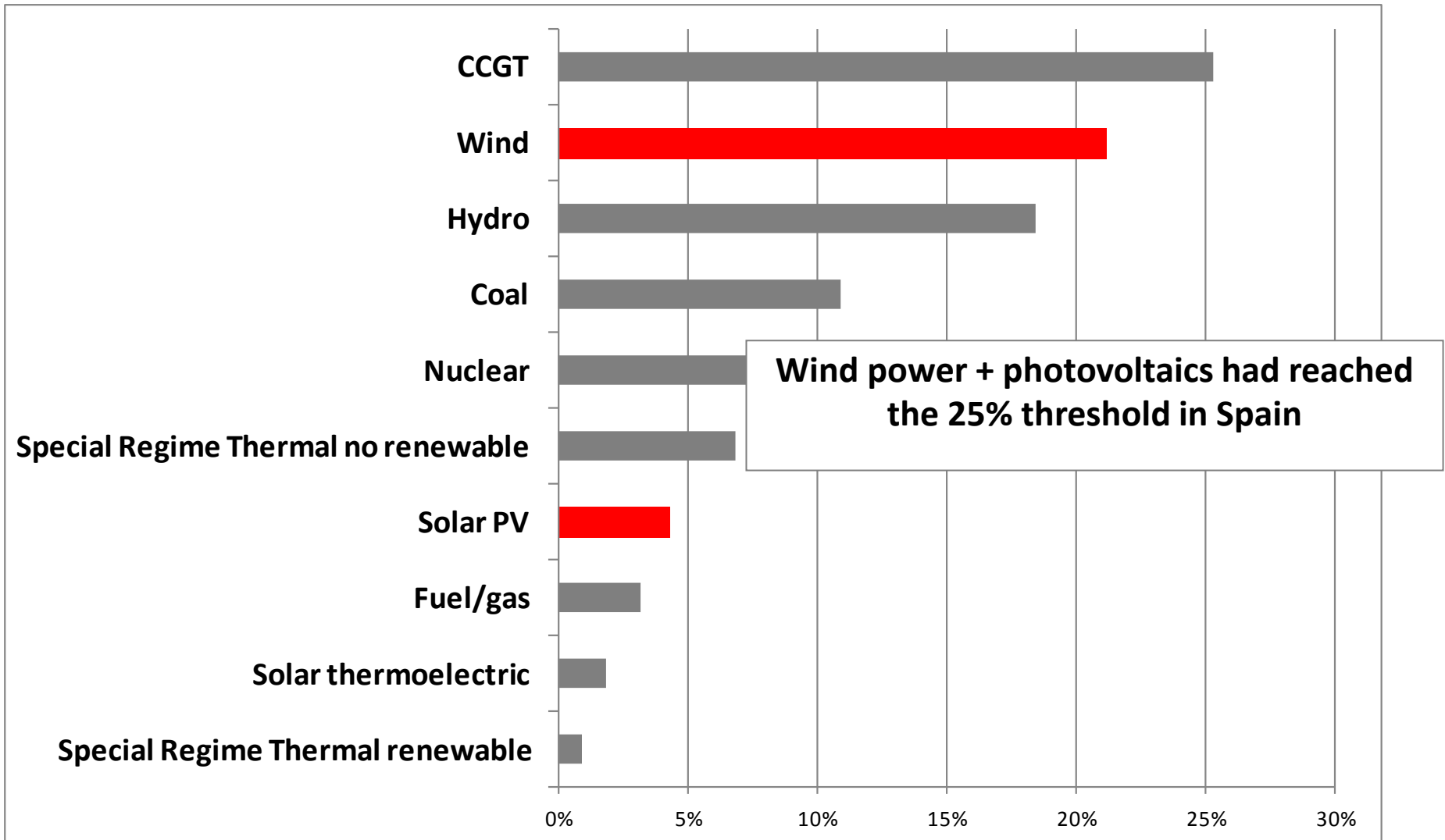
Price arbitrage - charging during off peak hours and discharging at peak hours



2. Conventional approach to storage: price arbitrage

Wind power has grown dramatically in Spain over the past years

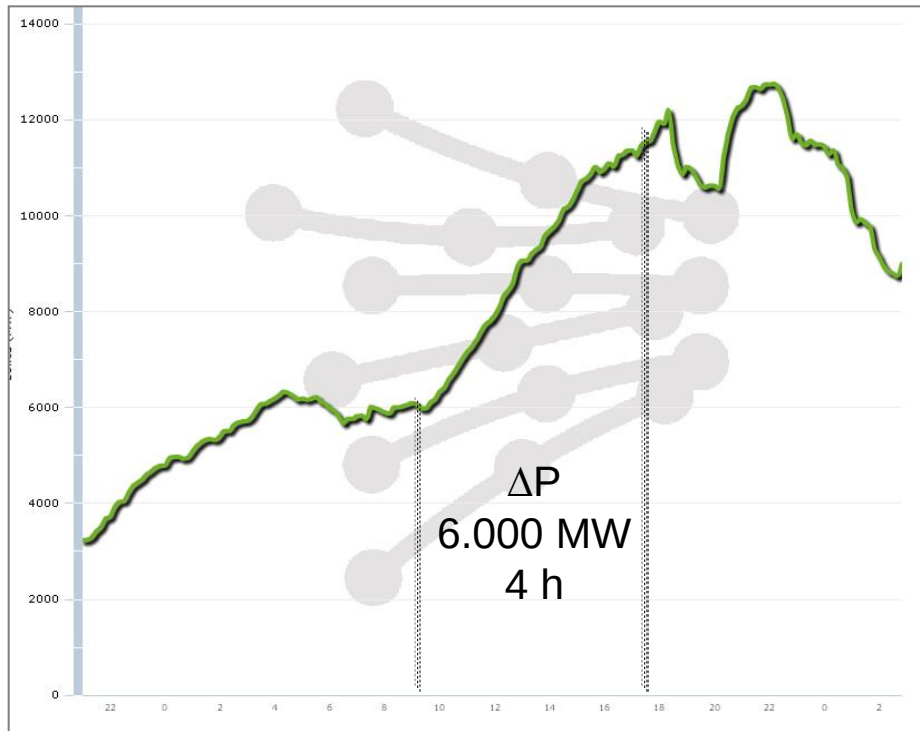
Capacity mix in Spain (%) 2012



2. Conventional approach to storage: price arbitrage

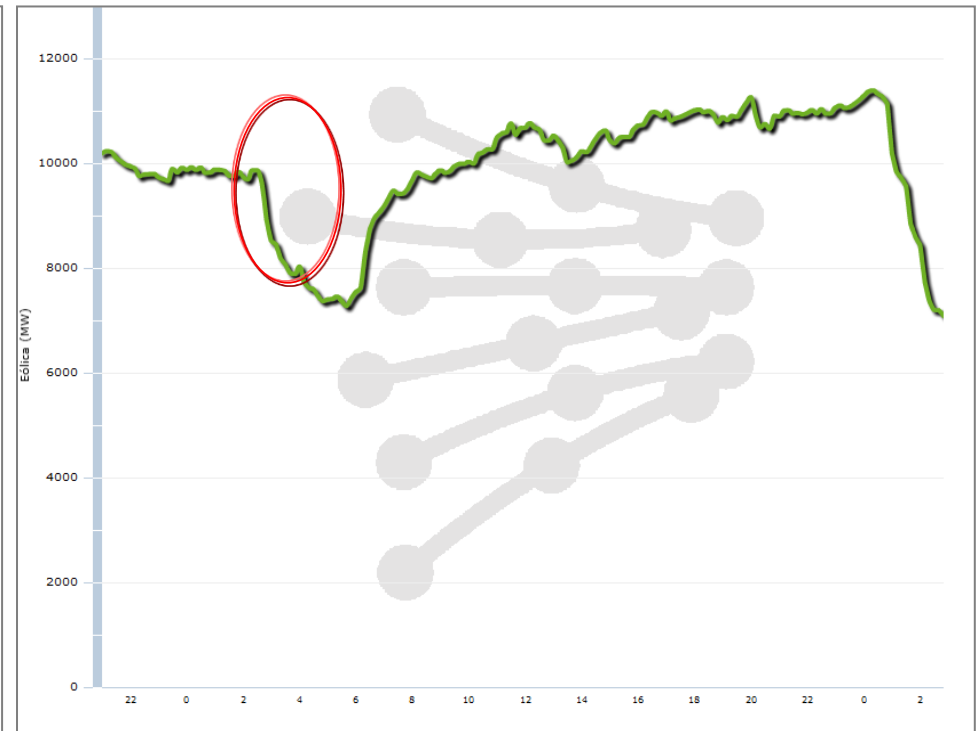
Now there is a Supply Side Management problem

Sometimes wind power grows a lot...



Wind power production in Spain 29.03.2010

...and has to be disconnected



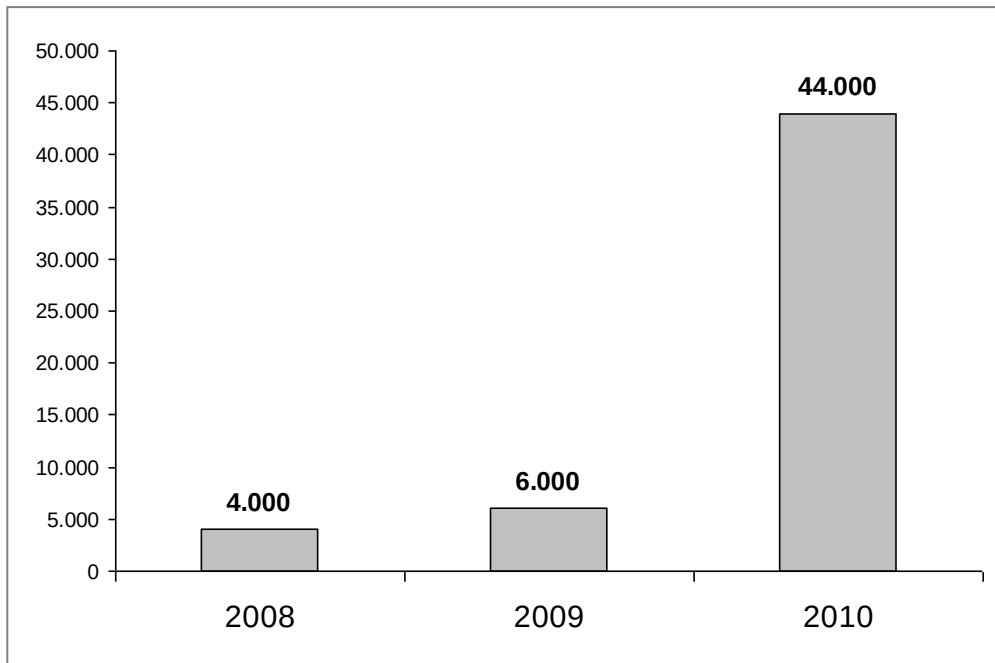
Wind power production in Spain 25.02.2010



2. Conventional approach to storage: price arbitrage

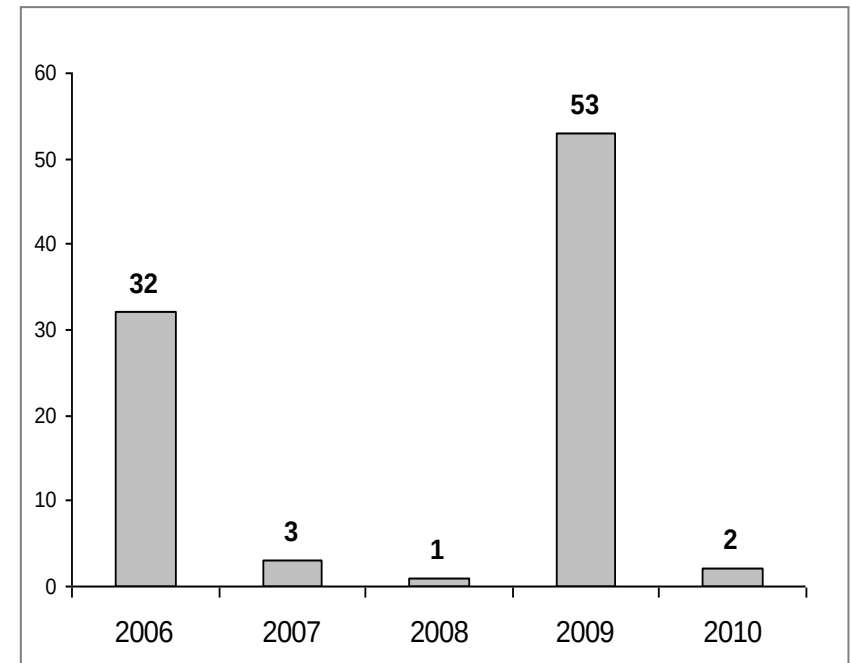
Supply “peak shaving” is the problem now

There is a lot of energy wasted...



Energy lost due to disconnection of wind
(MWh) in Spain

...but other fuels also suffer



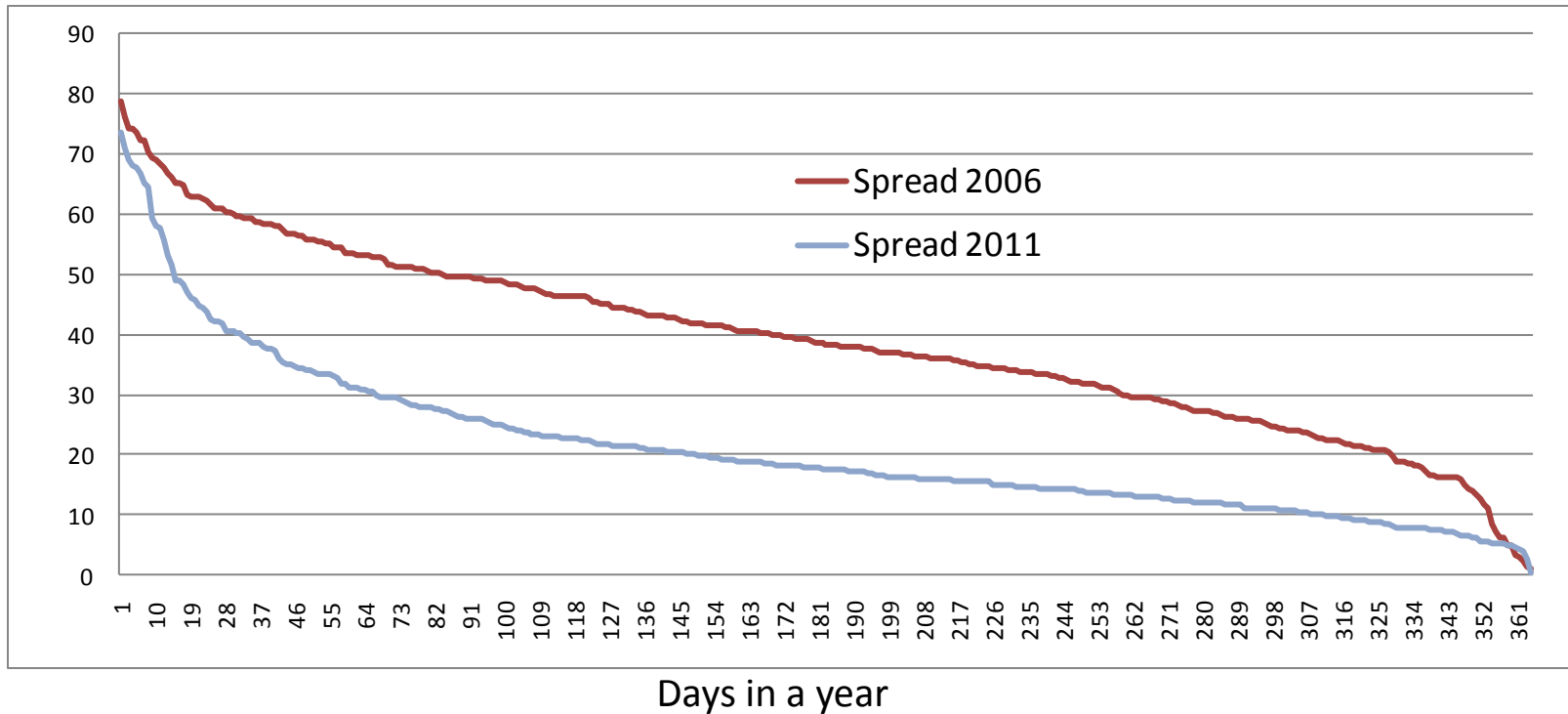
Thermal units disconnected to allow wind
production in Spain



2. Conventional approach to storage: price arbitrage

Peak and off Peak prices are getting closer: the spread for price arbitrage is reducing

**Spread (€/MWh) in Spain
(Peak - Off peak Prices)**



- Excess of capacity
- The number of days with a spread good enough to recover costs is reducing
- This puts more pressure on the economics of price arbitrage



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3. New approach: grid services and reserves

Threats and opportunities for energy storage

Energy storage point of view



RES have
reduced energy
spreads

Grid Services

Need for **quicker**
and **longer**
reserves

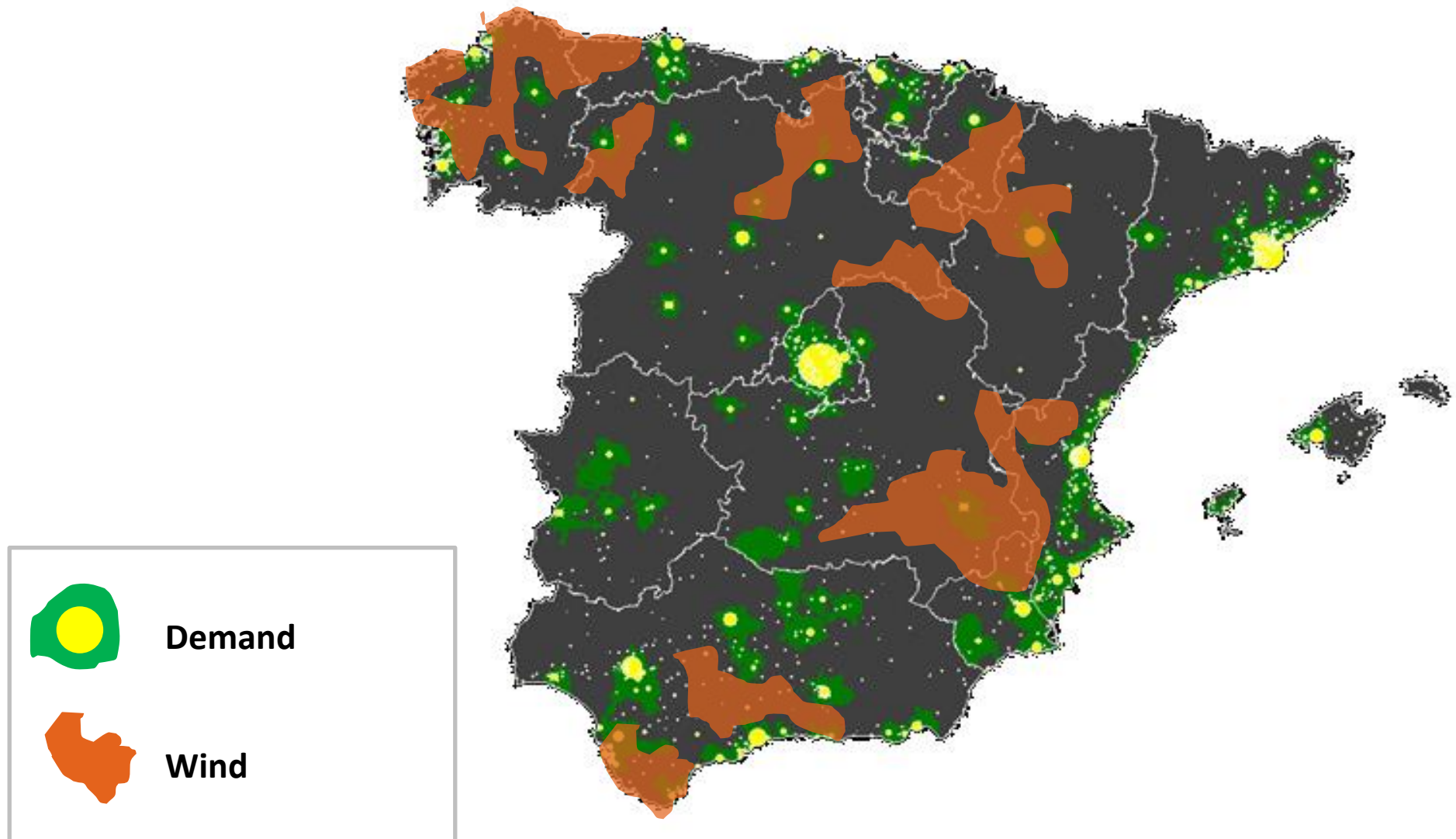


Intermittent
incontrollable
generating
capacity



3. New approach: grid services and reserves

Wind farms are far away from demand: more grids are needed

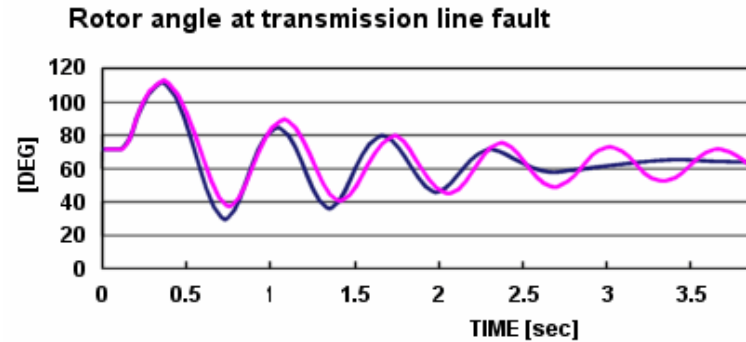


3. New approach: grid services and reserves

Services to transmission & distribution



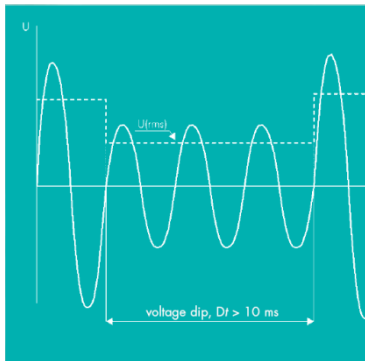
Investment deferral



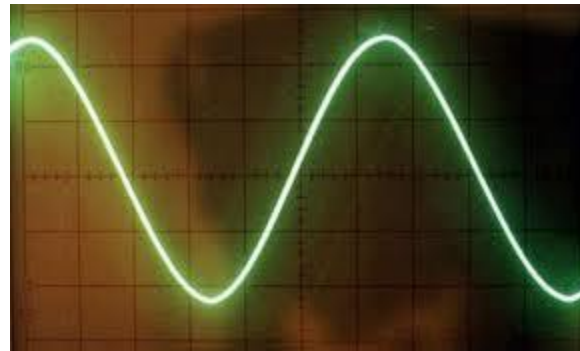
Grid stability



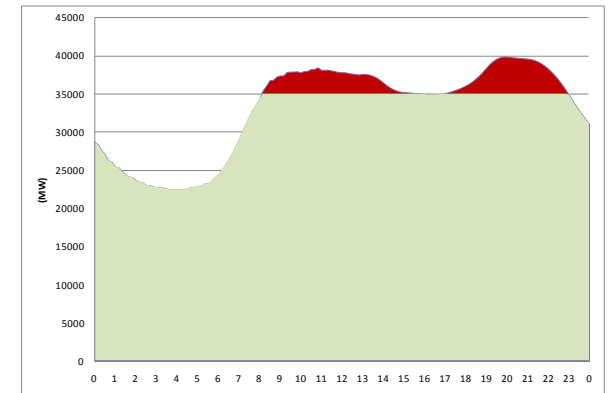
Contingency support



Voltage control



Frequency control



Charge management

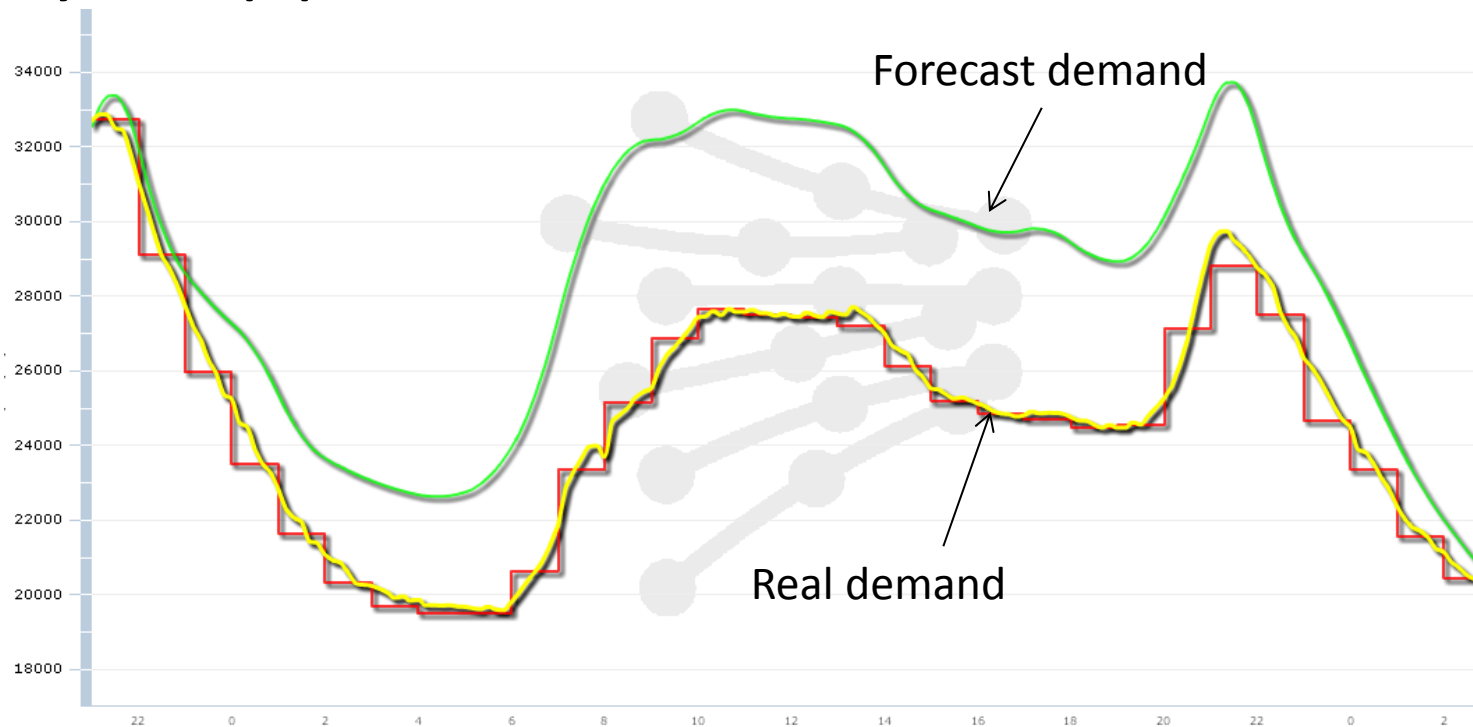
But all these services deserve a proper retribution



3. New approach: grid services and reserves

Need for fast and longer reserves

1. Future loads cannot be perfectly predicted
2. Generator outputs can vary substantially in different time frames
3. Power system equipment can fail without notice



Operating reserves: real power capability that can be given (upward) or taken (downward) to assist in generation and load balance and frequency control



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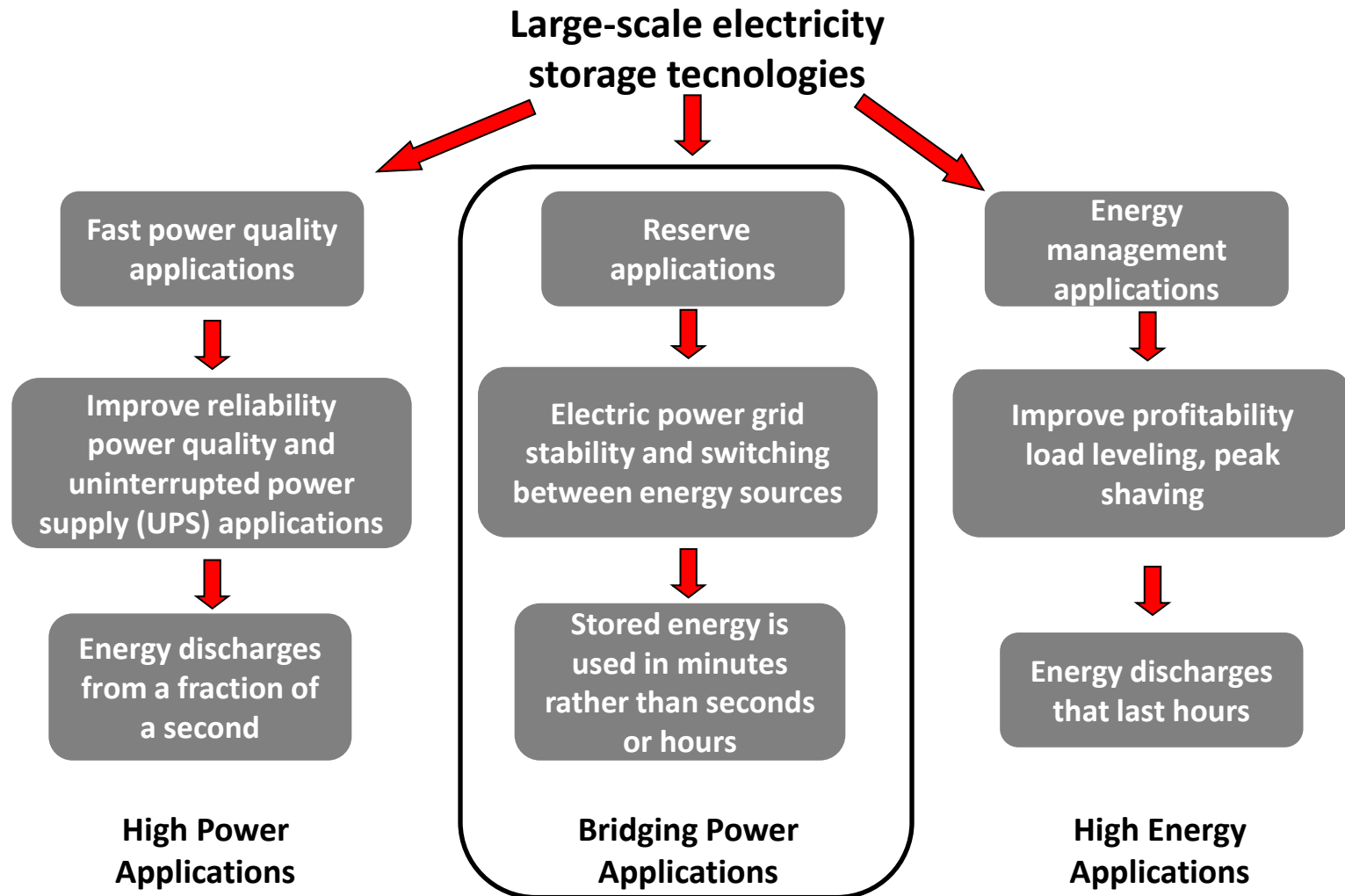
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4. Redox flow batteries are a promising option

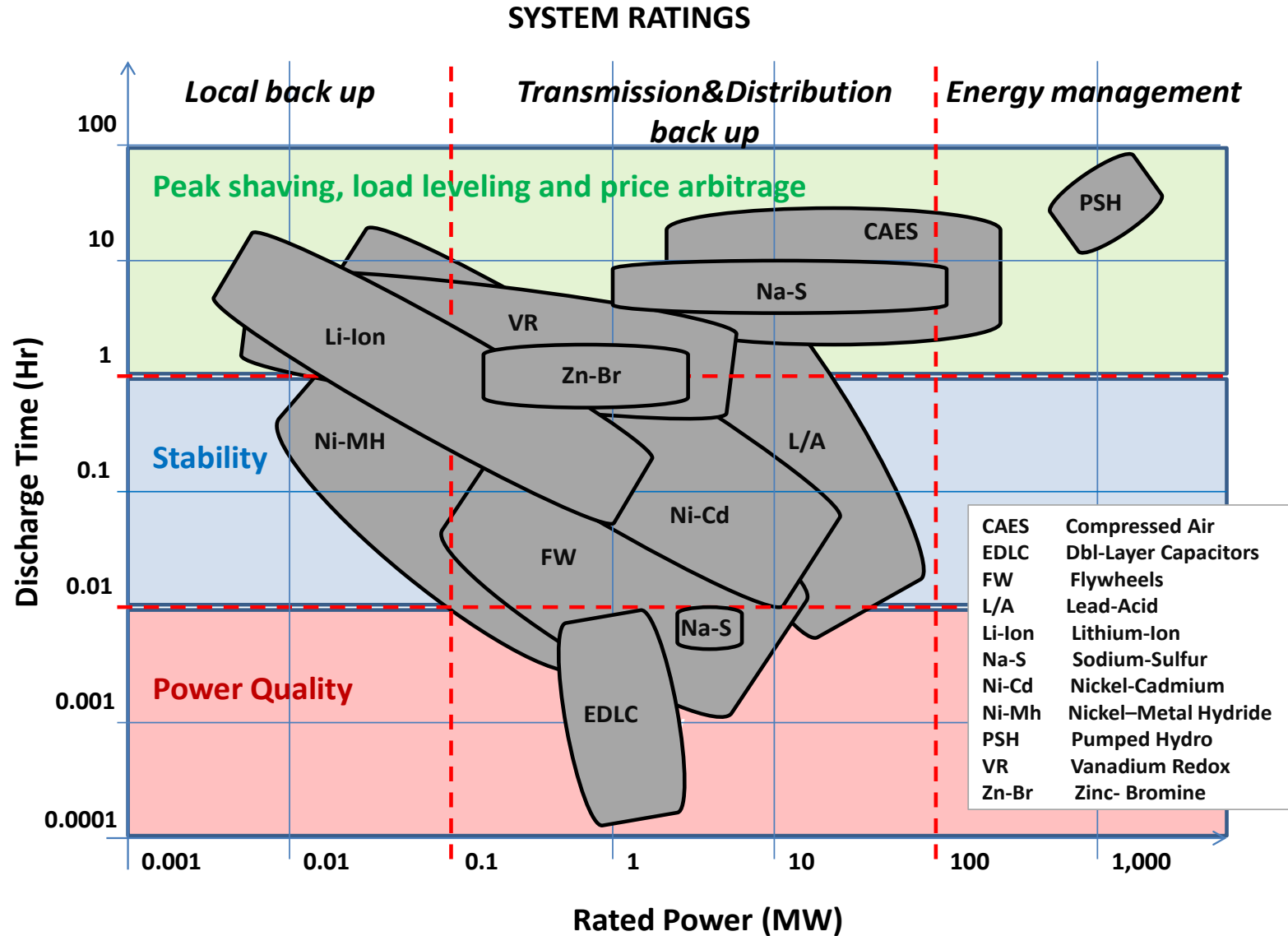
Different needs imply different technologies and approaches

What do you need storage for...?



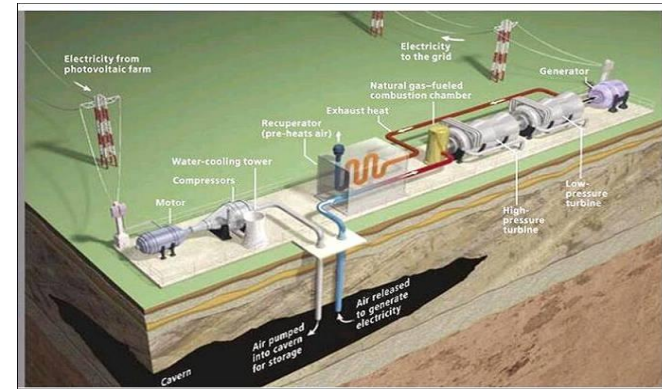
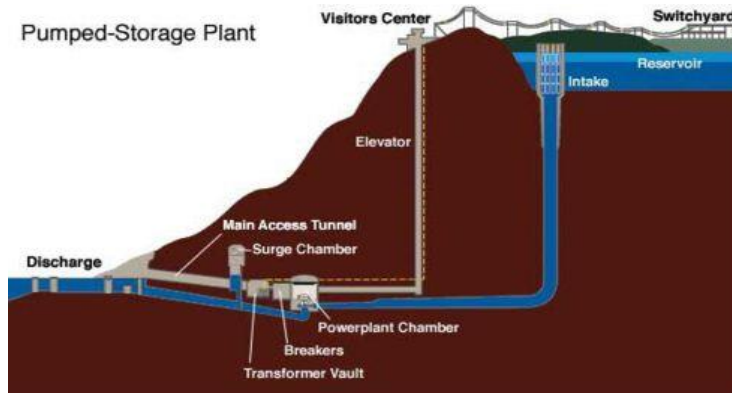
4. Redox flow batteries are a promising option

Power vs Capacity



4. Redox flow batteries are a promising option

Pumping and Compressed Air are for the lucky ones



Pumping hydro and Compressed Air Energy Storage fall apart from other technologies:

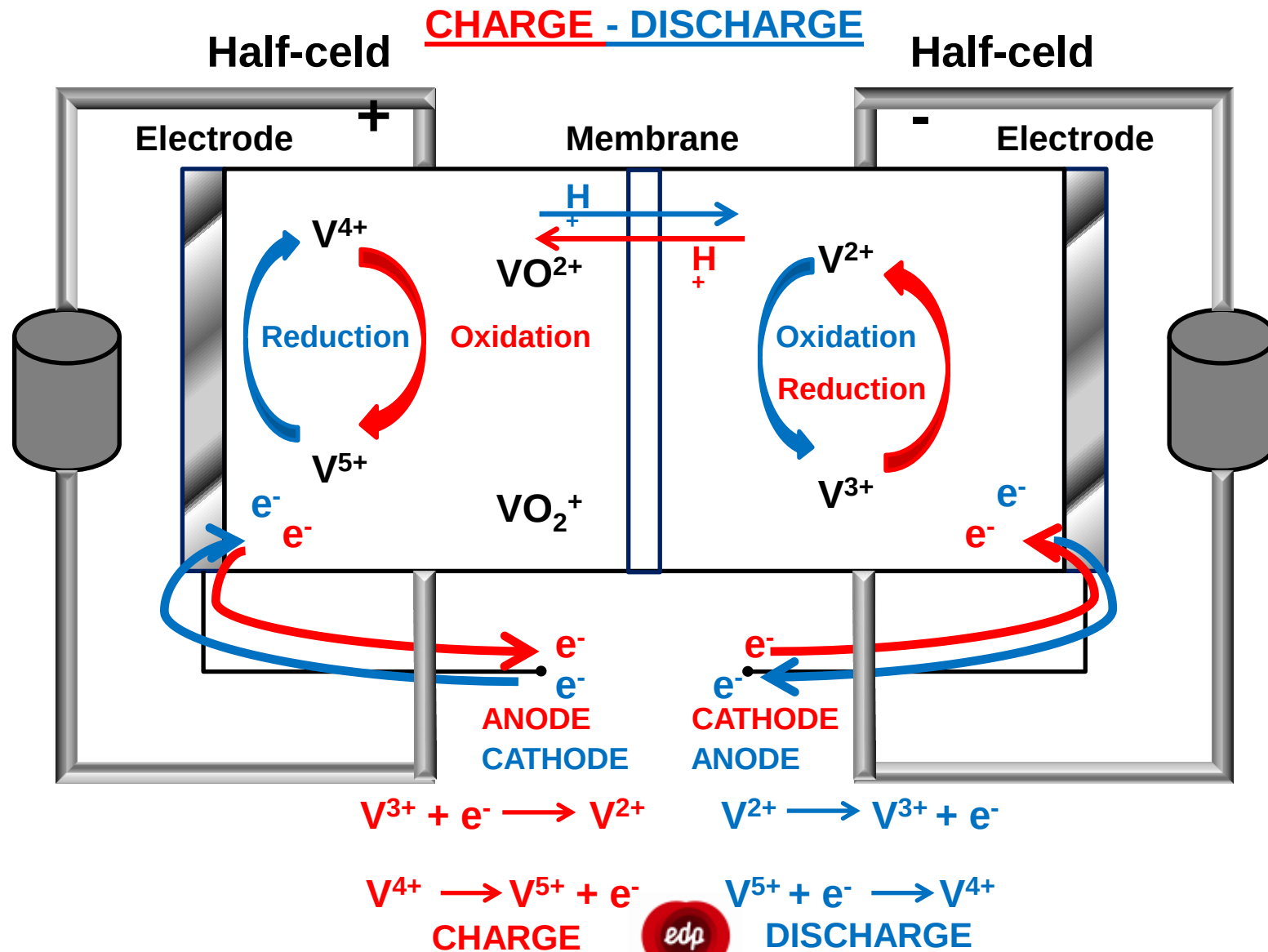
- Heavy reliance on geology (site dependent)
(Portugal, Austria, Switzerland, Slovenia)
- Massive capital costs and long comissioning periods
- Mature technology
- Enviromental issues involved

Other technologies are needed to address the storage problem from a distributed & site-independent approach



4. Redox flow batteries are a promising option

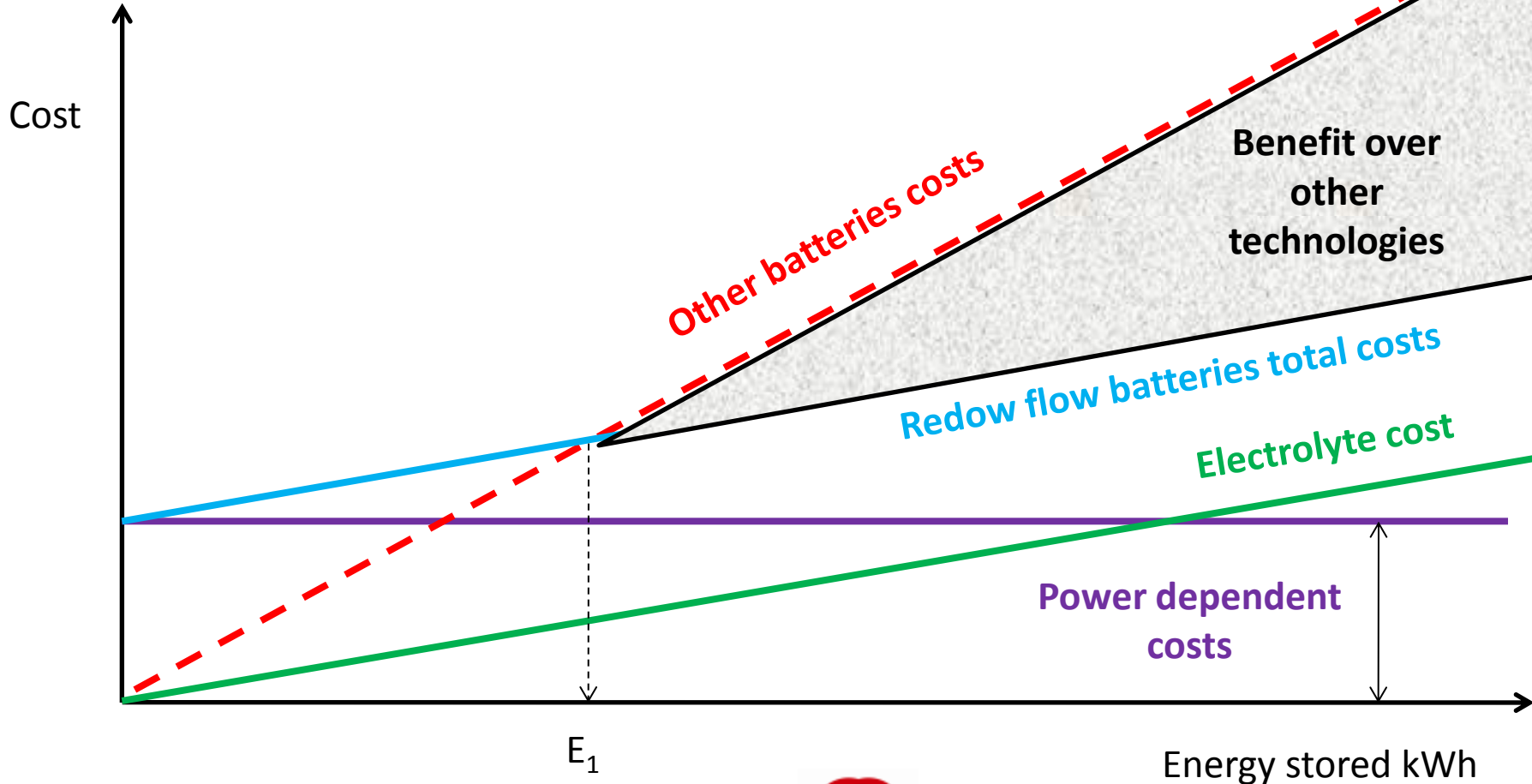
Redox flow battery



4. Redox flow batteries are a promising option

Capacity costs and power costs are independent

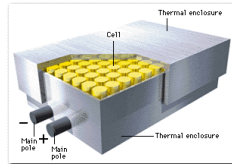
Redox Flow Batteries show better cost performance for large systems



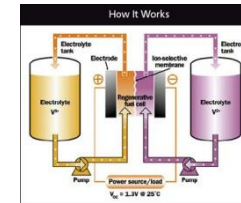
4. Redox flow batteries are a promising option

Redox flow batteries show more opportunities to bring costs down

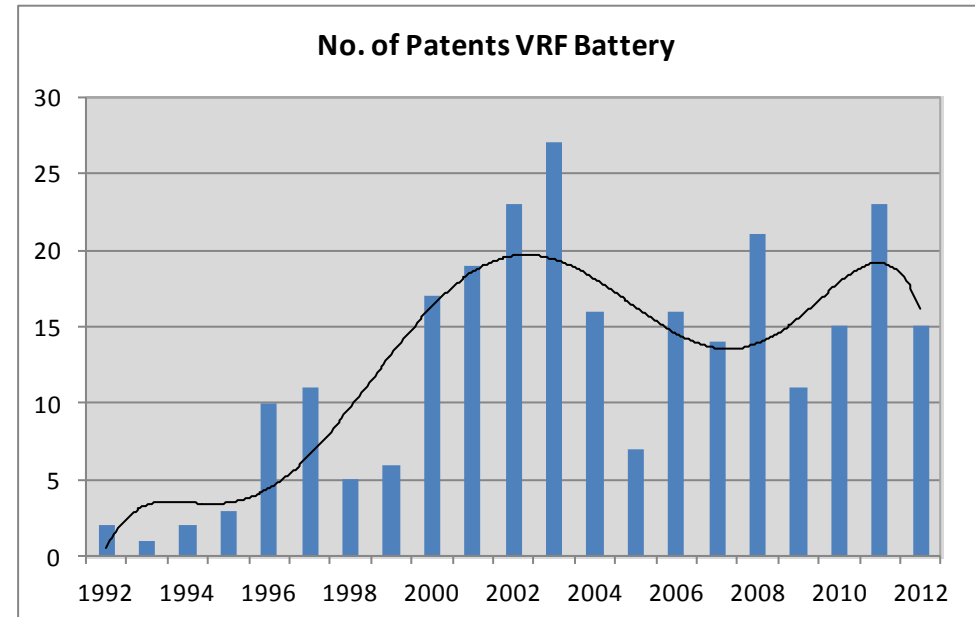
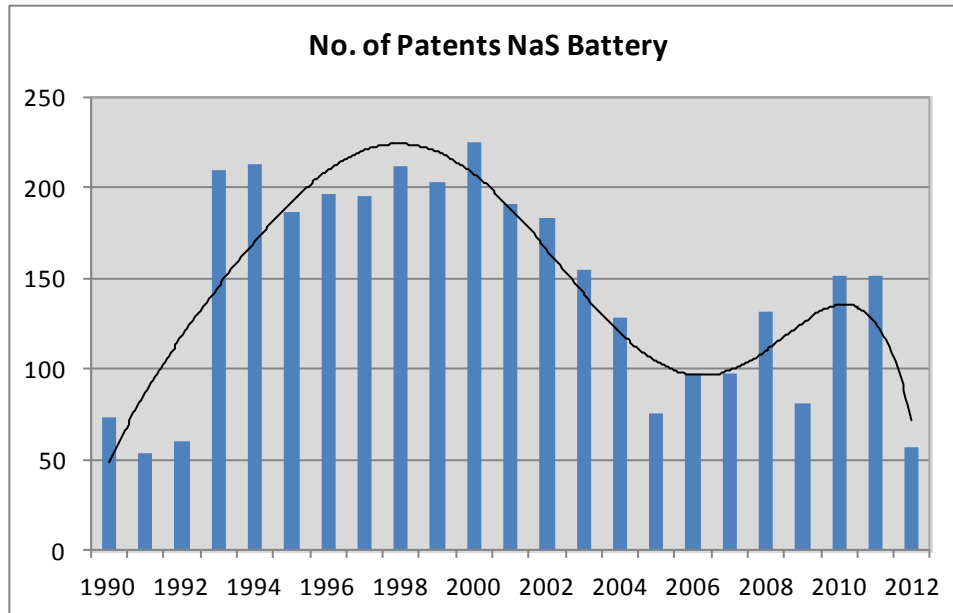
Tendency of published patents shows better perspective for Redox Flow Batteries



NaS battery



VRF battery



From a fire incident in 2011 NGK has asked nearly every customer to suspend the use of NAS batteries or restrict its use



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5. Redox2015 project

Consortium and funding



- Objective: develop a Vanadium Redox Flow Battery by spanish partners
- Budget :2,7 M €
- Project length: 27 months (2011-2013)

Funded by the Economy and Competitiveness Ministry of Spain with FEDER funding from the European Commission

(IPT-2011-1690-900000)

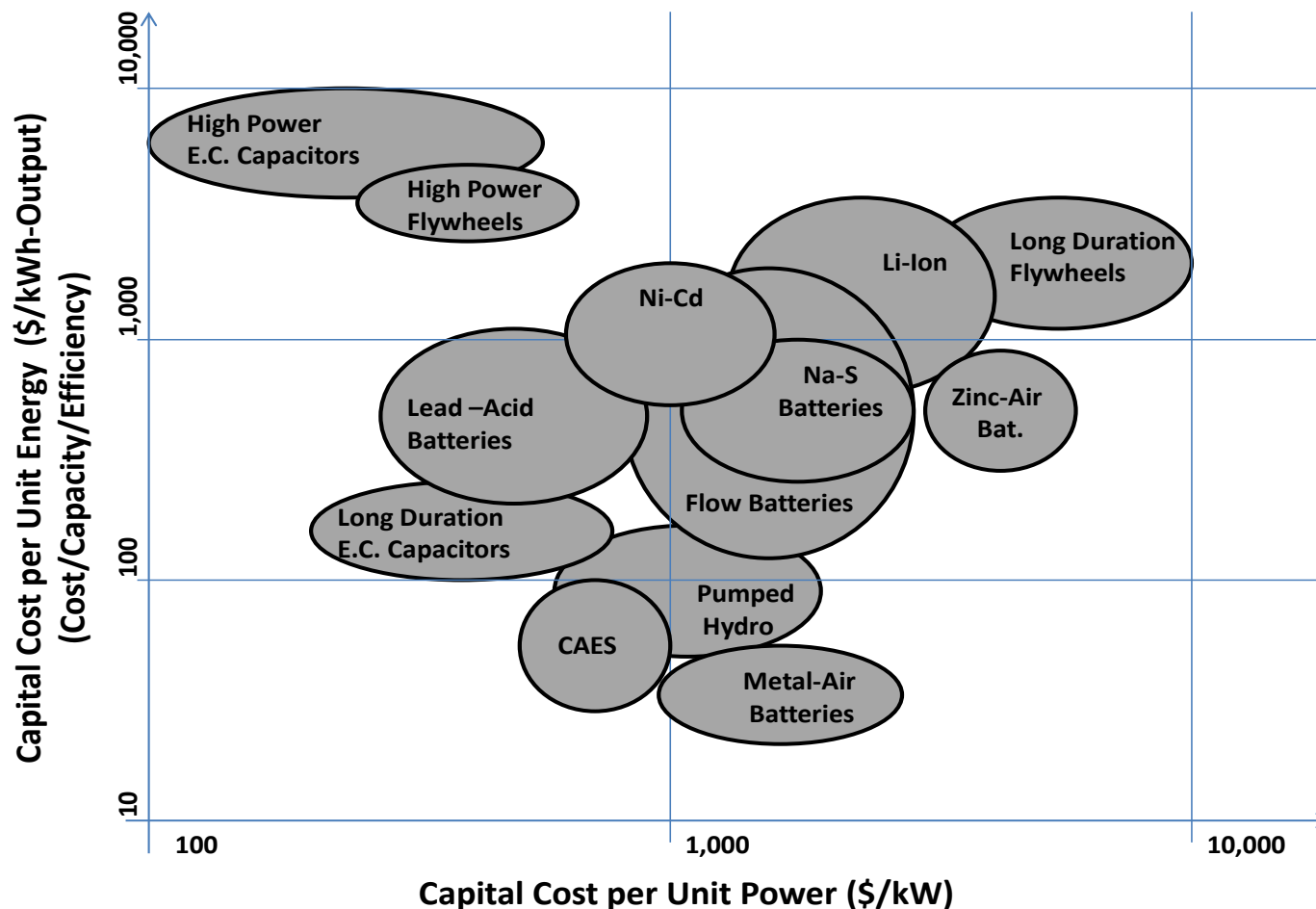


redOx 2015



5. Redox2015 project

Questions to answer by the project: metrics to evaluate



Cost metrics

- **capex** in €/kW and €/kWh

Performance metrics

- **real efficiency**
- **ciclability**
stability
- **calendar and cycle lives**



5. Redox2015 project

Findings so far



	Service	Required power		Required discharge time	Required response time	Frequency of use	MV connection					LV connection			
							A	B	C	D	E	F	G	H	I
Distribution	Capacity support	MV	500 kW – some MW	2 – 10 h	Minutes	Occasional (peaks)	●	●	●	●	●	●	●	●	●
		LV	hundreds kW												
	Local voltage control	MV	100 kW – some MW	2 – 10 h	Minutes	Occasional (peaks)	■	●	●	●	●	●	●	●	●
		LV	10 kW – hundreds kW												
	Contingency support	MV	100 kW – some MW	2 – 10 h	Minutes	Unusual	●	●	●	●	●	●	●	●	●
		LV	10 kW – hundreds kW												
	Island mode	MV	100 kW – 10 MW	4 – 10 h	20 ms	Unusual	○	○	○	○	○	○	○	○	○
		LV	10 kW – 1 MW												
	Reactive compensation	hundreds kVar – some MVar.			Minutes	Daily	○	○	○	○	○				
Generation	Peak capacity	More than 1 MW		1 -10 h	Minutes	Daily	■	■	■	■	■	■	■	■	■
	Local generation support	More than 1 MW		variable	Variable	Variable	□	□	□	□	□	□	□	□	□
	Services to DER	hundreds kW – 2 MW		20 min. – 1h30	15 30 seconds	Continual	●	●	●	●	●			●	
Transpo rting	Frequency control	More than 1 MW		20 min. – 1h30	15 – 30 seconds	Continual	●	●	●	●	●				
	Frecuency stability on island	More than 1 MW		Tens of seconds	Less than 1 second	Unusual	■	■	■	■	■				
Consumer. Managing peak demand		0,5 – 10 MW		1 -10 h	Minutes	Daily			●		●		●	●	●

■	Feasible	●	High value for storage (■ specific for island mode)
○	Inappropriate	○	Other applications are feasible (□ specific for island mode)



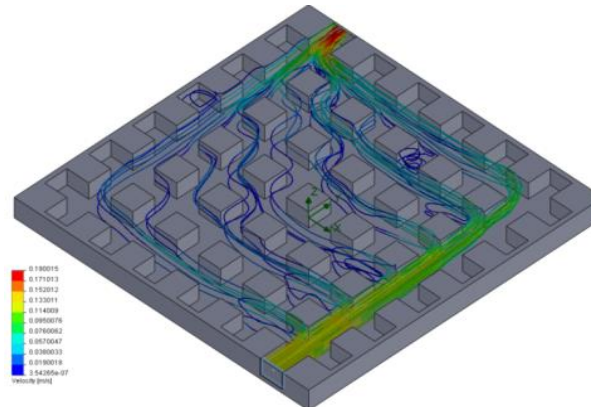
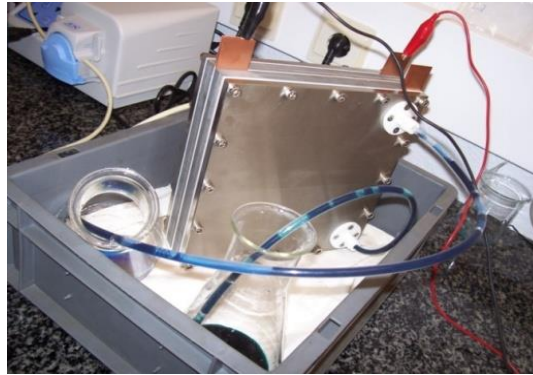
5. Redox2015 project

Findings so far



SIMPLE CELL DESIGN

1. Components screening
2. Cell design
3. Charger/Discharger prototipe
4. Flow dynamics modelling
5. Single cell optimizing tests
6. Stack development

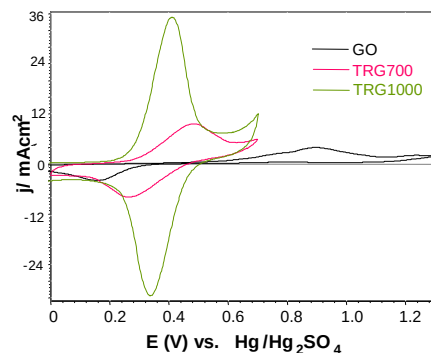
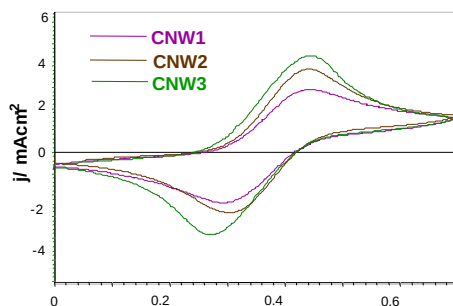
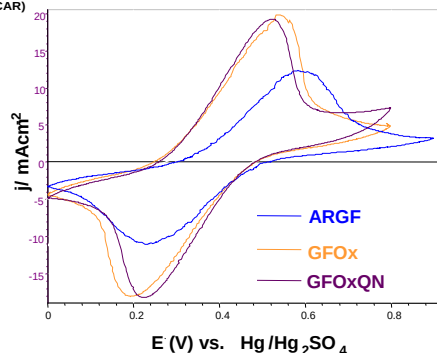


5. Redox2015 project

Findings so far



ELECTRODE ACTIVE MATERIALS SCREENING FOR VRFB

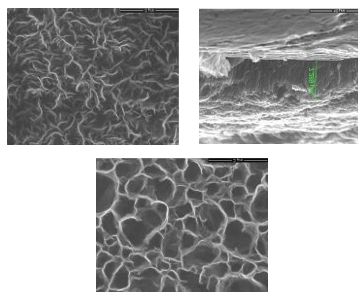
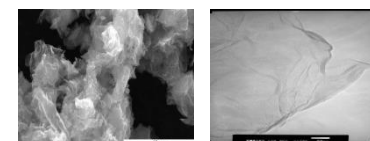


GO: Graphite Oxide

TRG700: GO reduced at 700 °C

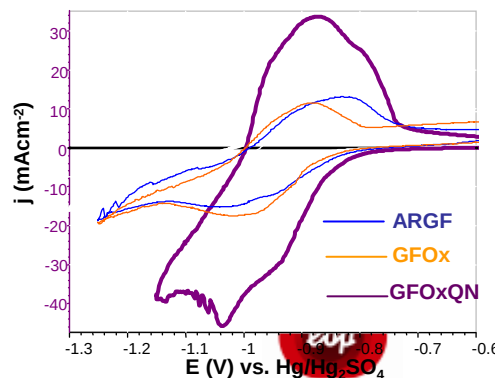
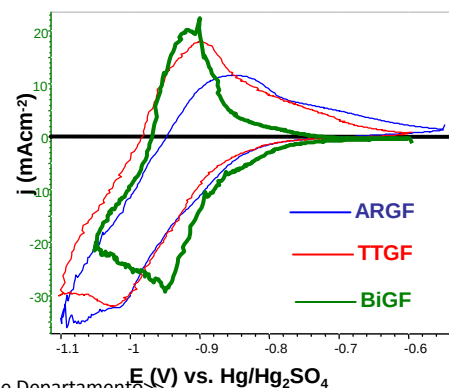
TRG1000: GO reduced at 1000 °C

GRAPHENES



CARBON NANOWALLS

**POSITIVE
HALF-CELLD (V^{4+/5+})**



NEGATIVE HALF-CELLD (V^{2+/3+})

ARGF: Felt untreated

TTGF: Thermal felt

BiGF: Modified felt using bismuth

GFOx: Electrochemically oxidized felt

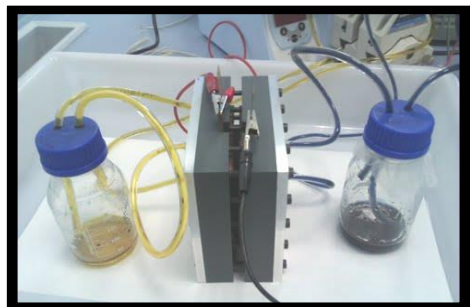
GFOxQN: chemically oxidized felt

5. Redox2015 project

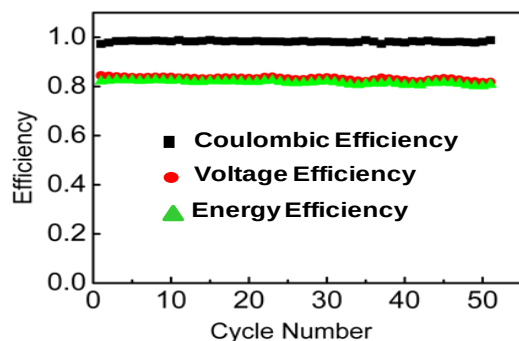
Findings so far



Institut de Recerca en Energia de Catalunya
Catalonia Institute for Energy Research



VRFB PROTOTYPE



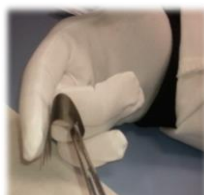
PAN commercial felt with standard electrolyte (3M H_2SO_4 and 1M $VOSO_4$)

Objective: High energy and power efficiency, long life and cyclability (excellent > 30,000), 5 times the charge density of the typical solutions

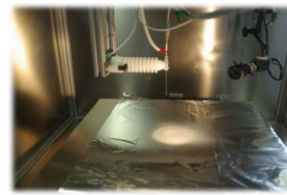


Electrodes development

Synthesis

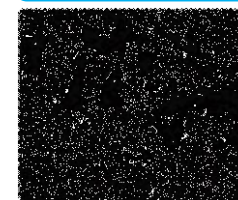


PAN based flexible nanofiber



Electrospinning

Functionalization

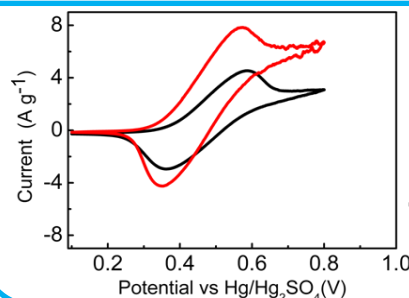


Graphene nanoparticles



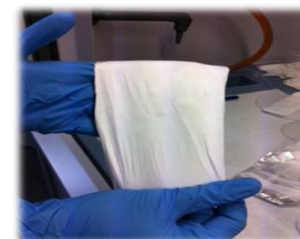
PAN commercial felt

High energy density electrolyte



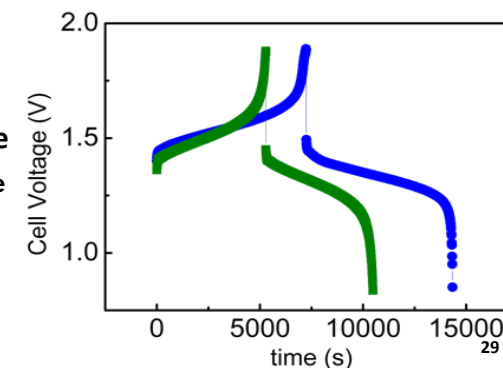
Polymeric additives in aqueous electrolytes

Innovative membranes



Use of additives to improve the electrolyte

- standard electrolyte
- additive-containing electrolyte



5. Redox2015 project

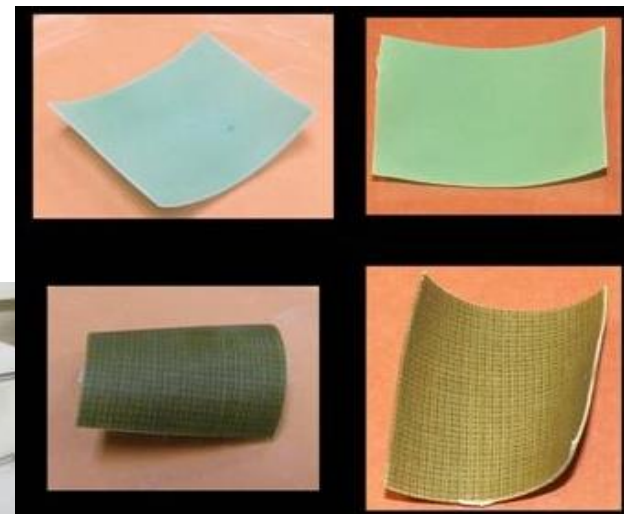
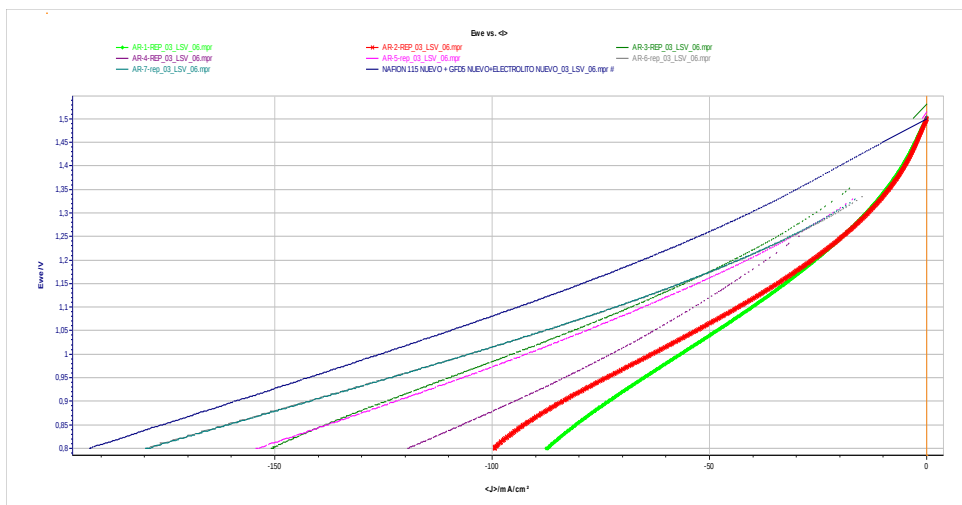
Findings so far



MEMBRANES FOR REDOX FLOW BATTERIES

- Objective: To minimize vanadium crossover through the membrane by surface plasma activation .
- These treatments cause changes in the polymer material:
 - Inversion of surface polarity
 - Surface crosslinking

tecna^{lia} Inspiring Business



REDOX FLOW BATTERY SINGLE CELL

- Single cell set-up to test new materials in flow battery operation
- Development of testing protocols for membranes and electrodes



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6. Conclusions



- **Energy storage for price arbitrage, at the current state of art and current prices, is not competitive**
- But the increase of penetration of wind power implies different conditions to take into account
- **Reserve markets are a good opportunity for energy storage economics**
- Where pumping is not an option, other applications of distributed energy storage are necessary
- **R&D projects like REDOX2015 are needed to meet performance and cost requirements**





**Thanks for your
attention!**

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